



Society of Petroleum Engineers

SPE-226667-MS

First Successful Application of CT Sand Jetting and Multi-Stage Proppant Fracturing on Horizontal Well in Carbonate Reservoirs in Iraq

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This paper was prepared for presentation at the SPE International Hydraulic Fracturing Technology Conference and Exhibition held in Muscat, Sultanate of Oman, 23 - 25 September 2025.

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Abstract

The ultra-low permeability carbonate S Formation (permeability <0.1 mD) in Iraq's X Oilfield accounts for 23% of total reserves, yet contributes only 0.3% of total production due to ineffective conventional development. Through pilot trials, 38 horizontal wells utilizing open-hole multi-stage proppant fracturing achieved an average production of 1,200 BOPD per well. Building on this success, a pioneering casing-cemented well integrating coiled tubing (CT) sand jetting with multi-stage proppant fracturing to achieve more frac stages was implemented in September 2024—the first application of its kind in Iraq. This paper analyzes the pilot well (H1X) to demonstrate the technical viability of this approach for ultra-low permeability carbonate reservoirs.

To ensure the success of this campaign, a holistic technical study including geology evaluation, reservoir performance analysis, drilling trajectory design, completion and fracturing technique design have been carried out based on principle of "geology-engineering integration". This paper will present a comprehensive illustration including technical design (CT perforation, main completion-fracturing technique, total scale optimization, fracturing fluid, proppant), job execution and post-frac production performance for this successful campaign.

Through holistic design, the pilot well(horizontal length: 1536 m) was optimized with 29-stage CT sand jetting & proppant fracturing, with a total fluid volume of 10099 m³ and a total sand volume of 1134.6 m³. The well has achieved a production breakthrough of 2227 BOPD (28/64" choke) after frac. Compared with other open hole fractured horizontal wells, the estimated production index in is around 4.2 BOPD/psi, 2-4 times of other wells.

As the Middle East's first cased-hole multi-stage fracturing in ultra-low permeability carbonates, this case study sheds light upon and provides a replicable template for unlocking similar challenging reservoirs.

Background

The Middle East is home to abundant oil and gas resources, with carbonate reservoirs playing a crucial role in global energy supply. However, ultra-low permeability carbonate reservoirs, such as the S Formation in Iraq's X Oilfield^[1], present significant challenges to effective development. The S Formation has a permeability of less than 0.1 mD, which makes conventional development methods ineffective. Despite accounting for 23% of the total reserves in the X Oilfield, its contribution to total production is only 0.3%, highlighting the urgency to find a viable development approach.

S formation is a limestone reservoir with unique petrophysics, i.e., medium to high porosity (15-20%) but low permeability (0.05-1.3 mD), poor connectivity between pore throats. Before proppant fracturing, there were six wells including 4 vertical wells and 2 horizontal wells in S formation. As the conventional routines for the reservoir stimulation in carbonate formation, matrix acidizing(Y2011-2012) as well as acid fracturing (Y2012-2013) were implemented on these wells in S formation in the early stage. After the treatments, the initial production was high, indicating a good stimulation effect. However, the production declined rapidly by 50%-65% in both the vertical and horizontal wells within 2-3 months, resulting in intermittent production and 3 wells being shut in. Due to the poor petrophysical conditions and limited stimulation degree, the stimulation effect was not satisfactory and has not achieved the production target by conventional methods.

From Y2015, proppant fracturing on Sadi was initiated. By a serious of technical study and demonstration, the first vertical Well Pilot test was executed in Y2016, followed by the second well in Y2018. The first hydraulically fractured horizontal well was deployed and successfully executed in Y2019 and . Since then, a serial of horizontal wells with multi-stage fracturing were deployed. By the end of Y2023, 26 horizontal wells based on "open-hole packer and ball-activated sliding sleeve" multi-stage fracturing technique have been implemented, covering Type I, II, III and IV sweet spots areas. In Y2024, a novel technique- coiled tubing (CT) sand jetting with multi-stage proppant fracturing was proposed and successfully executed on casing-cemented well H1X in S formation.

This paper focuses on the pioneering casing-cemented well (H1X) in Iraq's X Oilfield, which integrates coiled tubing (CT) sand jetting with multi-stage proppant fracturing. The primary objective is to demonstrate the technical viability of this approach for ultra-low permeability carbonate reservoirs through an in-depth analysis of the H1X pilot well. The scope includes geology evaluation, reservoir performance analysis, drilling trajectory design, completion and fracturing technique design, job execution, and post-frac production performance.

Wellbore structure overview

The drilling trajectory of the H1X well was carefully designed to intersect the most favorable reservoir zones. Considering the horizontal section length of 1,536 m, the trajectory was optimized to minimize drilling risks and maximize the contact area with the reservoir as shown in Fig.1. The design took into account the reservoir structure and formation pressure to ensure the wellbore stability and efficient reservoir drainage.

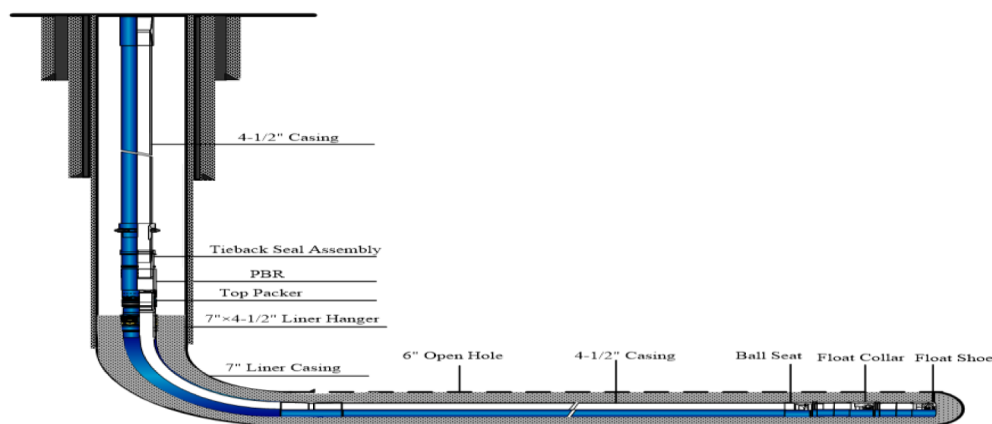


Figure 1—Diagram of wellbore structure of well H1X

The wellbore structure parameters are presented in [Table 1](#).

Table 1—Wellbore structure parameters of well H1X

Casing/ Tubing	Depth (m)	OD (in)	ID (in)	Wt (lb/ft)	Grade	Burst pressure (MPa)
Casing 1	155.0	20	19.124	94.0	K-55	14.55
Casing 2	1425.0	13- ³ / ₈	12.415	68.0	N-80	34.61
Casing 3	1989.0	9- ⁵ / ₈	8.681/8.535	47.0/53.5	L-80/T-95	47.37/64.88
Casing 4	2981.0	7	6.184	29.0	L-80	56.26
Tubing(tie back)	2782.0	4- ¹ / ₂	3.958	12.6	L-80	58.1
Casing (horizontal)	4565.0	4- ¹ / ₂	3.826	15.1	P-110	99.4

Holistic technical study framework

To ensure the success of the H1X pilot well campaign, a holistic technical study was conducted based on the principle of "geology-engineering integration"(Fig. 2). This framework emphasizes the close integration of geological understanding and engineering practices^[3], ensuring that each technical design step is aligned with the reservoir characteristics and development goals.

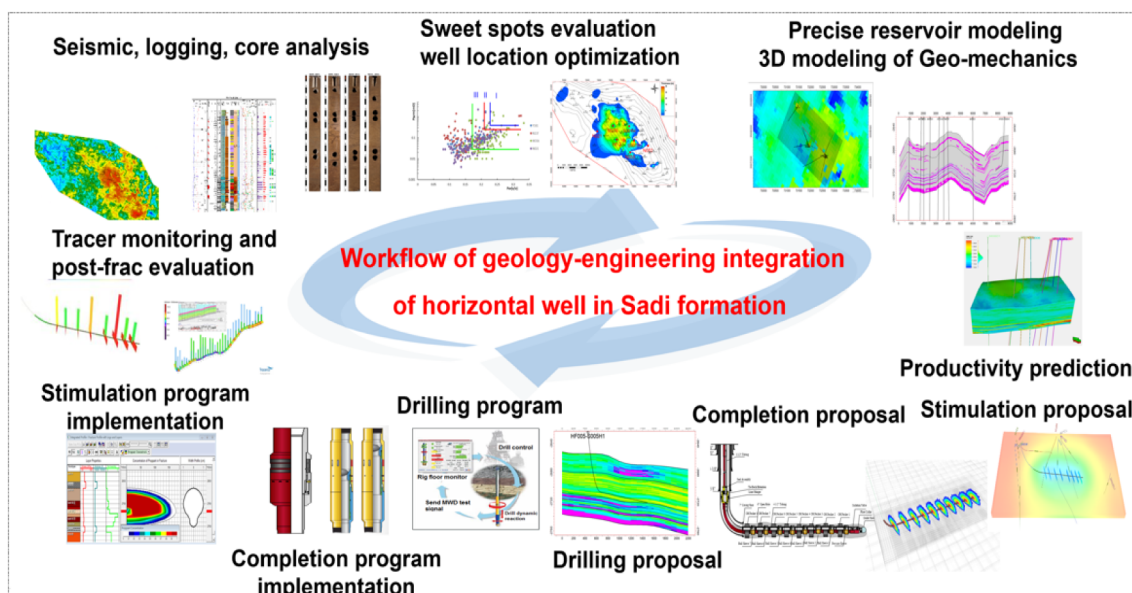


Figure 2—Workflow of "geology-engineering integration"^[7]

Perforation and fracturing technique design

CT Perforation

Hydrajetting Perforation(HJP) tools is used to perforate casing and cement. In the HJP process, abrasive fluid is pumped through a jet nozzle, converting the pressure energy to kinetic energy related to fluid velocity. According the industrial practice and study, the jetting velocity with the range of 130-200 m/s is required to penetrate through the casing and cement, aiming to create the communication holes with the formation. To ensure the successful and effective perforation, the recommended jetting velocity is at least 150m/s. For HJP technique, the most common nozzle size has been 3/16" (4.76mm) jets^[4], with jetting time of 10 minutes. The effective perforating is accomplished with 40/70 mesh silica sand, nominally about 100-120 kg/m³ concentration in lightly gelled fluid with a pumping rate of 0.65 m³/min.

Main completion-fracturing technique

CT sand-Jet Multi-Stage Fracturing Technique is based on the coiled tubing for frac stimulation with bottom-hole assembly composed of disconnect shear sub, centralizer, ball seat, sand-jet perforating sub with nozzles, isolation packer, etc. The abrasive jetting tool is used for perforation operations. This technique is used to conduct multistage fracturing of casing horizontal wells without wireline perforation gun. The fracturing fluid with proppant is pumped via the annulus between the coiled tubing and casing. After the first stage frac operation, then second stage is perforated and annular pumping is resumed to frac the second stage. This operation is repeated until all stages are completed. This is a continuous plug-less operation that eliminates the Run-in-hole (RIH) time during wireline perforation and plug setting. The procedure of the technique is illustrated as Fig. 3.

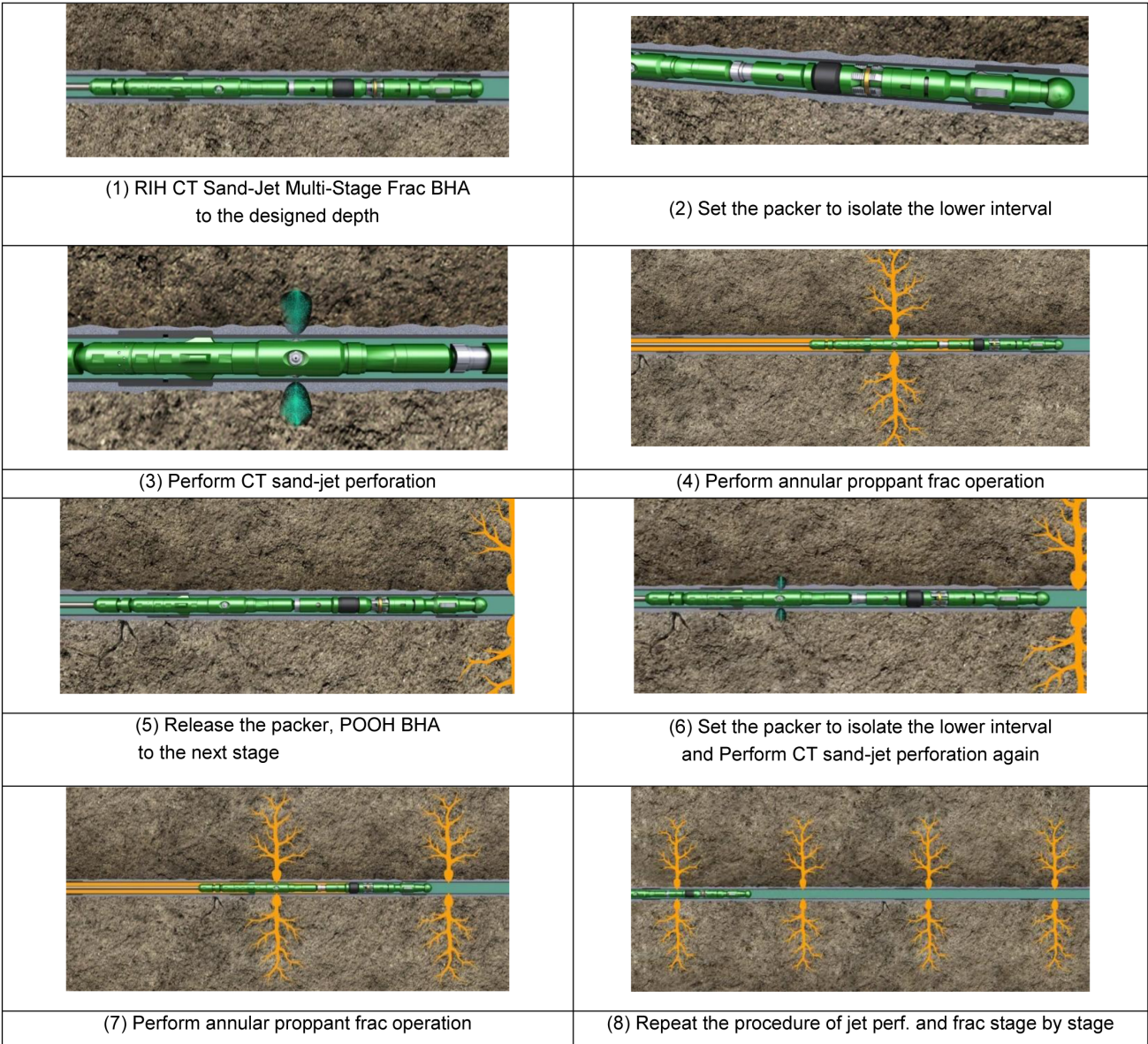


Figure 3—Procedures of CT sand-jet multi-stage fracturing technique

Total treatment scale optimization

Geology and Engineering Integration Modeling is utilized to predict the oil rate and cumulative oil under the condition of different stages to determine the optimum number of frac stages. The geology model is designed according to the actual well data, and predicted oil rate and cumulative oil production results are demonstrated as below.

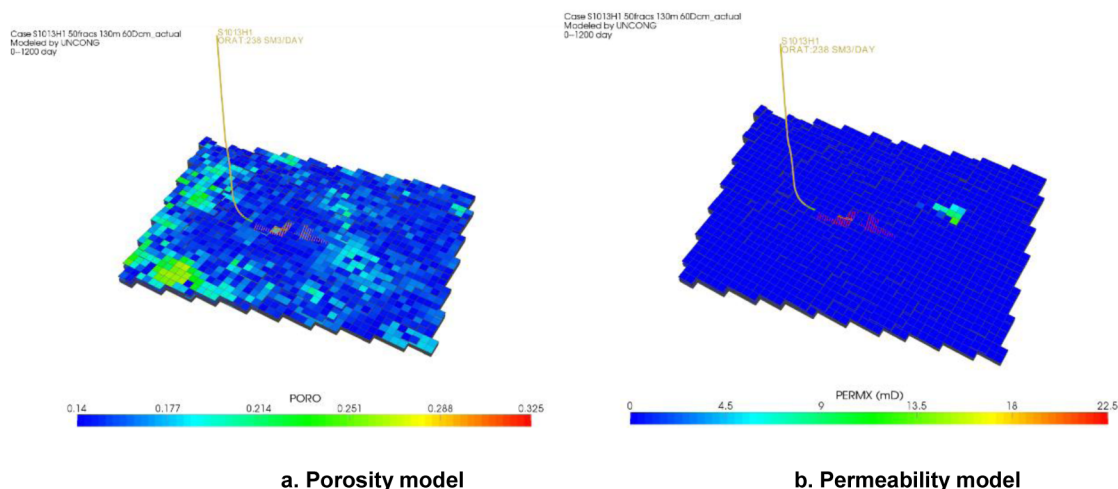


Figure 4—"Geology and engineering integration" model for well H1X

According to the accumulated oil production curve in Fig. 5, 29 stages is a turning point where the oil rate and accumulated oil production curves increasing slope become slower. Thus, the reasonable stages are optimized to be 29.

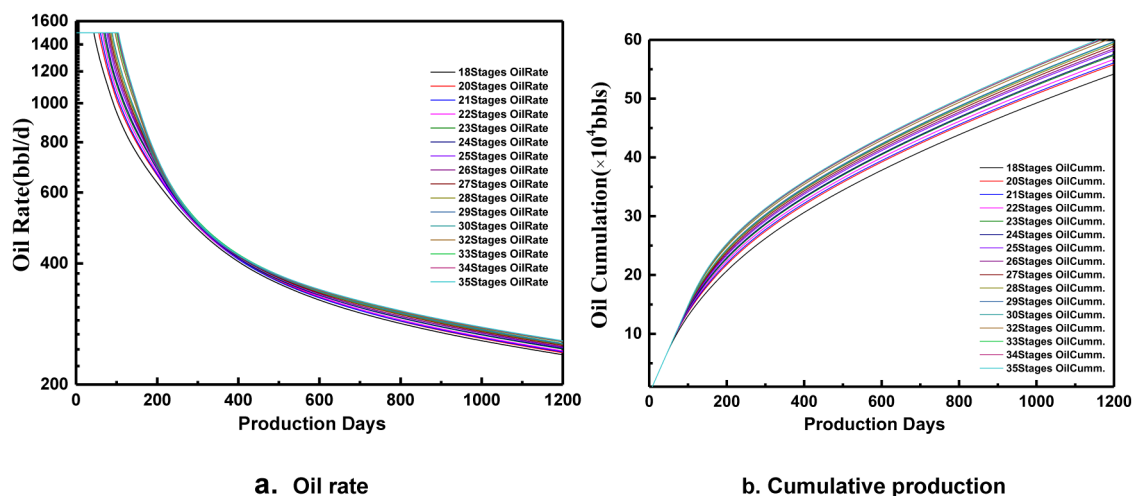


Figure 5—Oil production prediction for different fracturing stages

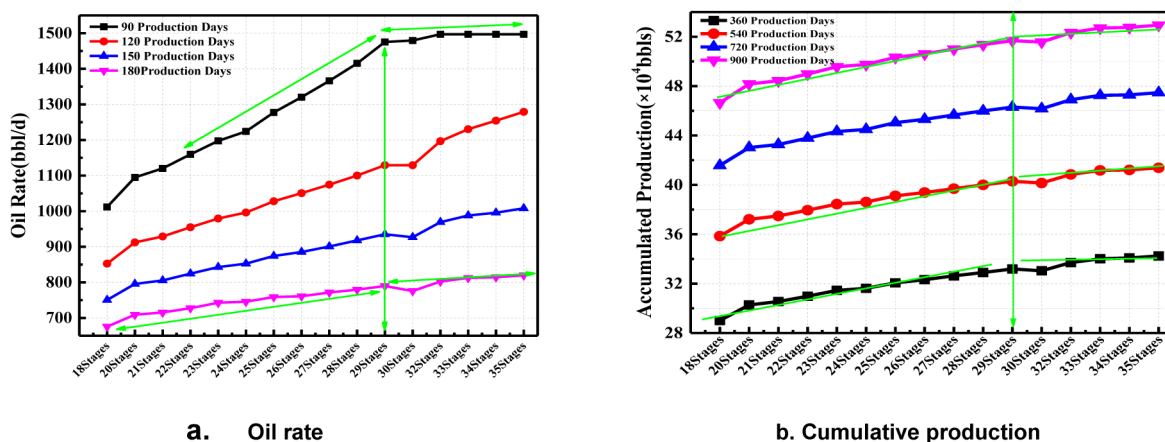


Figure 6—Oil rate and cumulative production of different stages for different time

The total scale of the fracturing treatment, including the total fluid volume and total sand volume, was optimized based on reservoir simulation results and field experience. For the H1X well, a total fluid volume of 9,975.7 m³ and a total sand volume of 1,105.0 m³ were determined. This optimization aimed to create sufficient fractures to connect the reservoir with the wellbore.

Fracturing fluid & proppant selection

A suitable fracturing fluid was selected based on the reservoir characteristics. The temperature of S formation is about 89°C. Based on previous experiences, 0.27% Hydroxypropyl Guar gum (HPG) fracturing fluid system is recommended. In combination with the experience of pilot test well, the pressure resistance of 52 MPa large size 20/40 mesh ceramsite is selected as proppant in all the pumping stages to increase the fracture conductivity.

Job execution

The job execution of the H1X well followed a strict operational procedure. The CT sand jetting perforation was carried out first, followed by the multi-stage proppant fracturing. Each fracturing stage was monitored in real-time to ensure that the fracturing parameters were within the designed range. The operation team closely coordinated with the design team to address any unexpected issues during the execution process, ensuring the smooth completion of the fracturing job.

Fig. 7 illustrates the operation curve of the 15th stage in coiled tubing (CT) dragging fracturing. It systematically breaks down the process into five sequential phases: CT perforation, Pad (slug), Slurry, Displacement, and Pressure decline monitoring. In the CT perforation phase, pressure fluctuations reflect the abrasive jetting action. As the process advances to the Pad phase, rising frac pumping rates drive up annulus pressure. The Slurry phase sees proppant injection, with the sand ratio climbing and pressures adjusting to proppant-laden fluid flow. During Displacement, proppant pumping ceases, rates reduce, and pressures start to decline. Finally, the Pressure decline monitoring phase features minimal pumping, with pressures trending downward to assess fracture closure and conductivity.

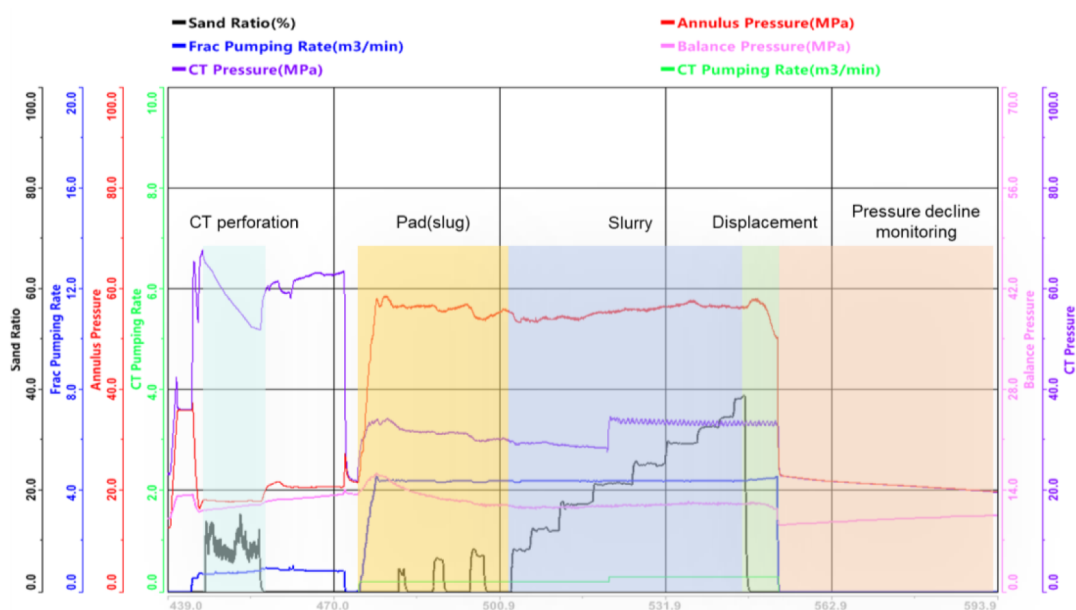


Figure 7— Operation curve for Stage No.15

Job execution results

The H1X pilot well was successfully optimized with 29-stage CT sand jetting & proppant fracturing, with a total fluid volume of 10,099 m³ and a total sand volume of 1,134.6 m³. The job execution went smoothly, with no major operational incidents, demonstrating the feasibility of the designed techniques and procedures.

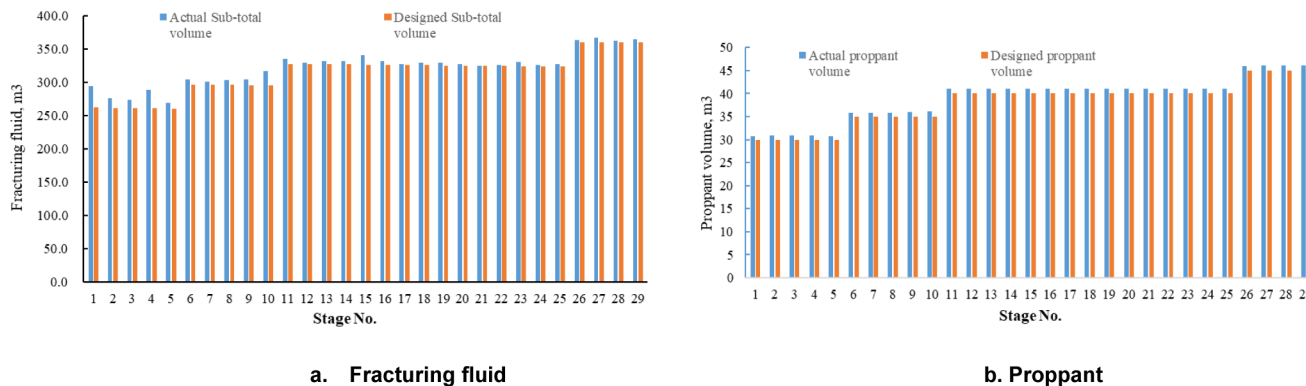


Figure 8—Comparison of fluid/proppant volume between actual treatment and design

Based on actual fracturing data, profile simulation for each stage is carried out and key fracture parameters are calibrated by net pressure matching for all 29 stages. Fig. 9 show the simulation and net pressure result for Stage No.15.

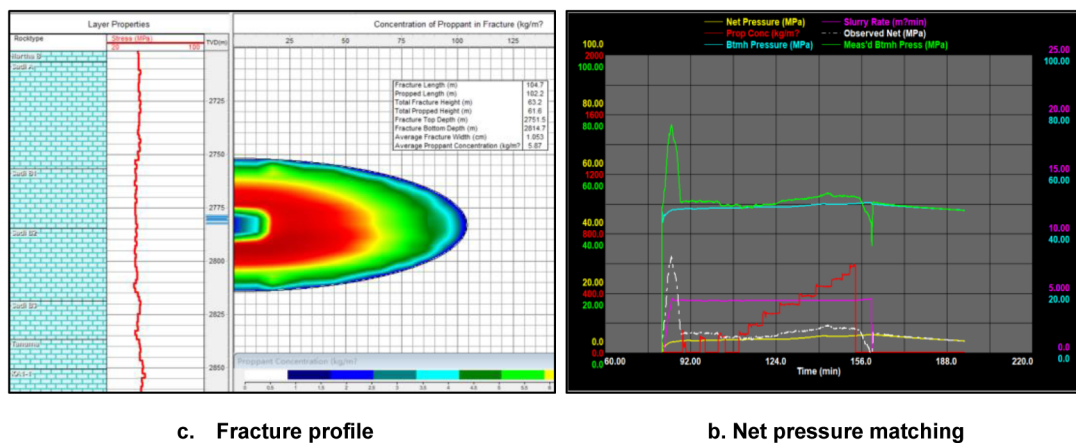


Figure 9—Fracture profile simulation and net pressure matching result for Stage No.15

Key fracture parameters for design and matching after frac (fracture half-length, propped half-length, average proppant concentration, average conductivity, maximum fracture height, and fracture top/bottom height) have been calibrated and listed in Table 2 (1-15 stages) and Table 3 (16-29 stages). Generally, the matching values are fairly close to the designed values. The average propped half-length and conductivity are 104.4 m and 701.5 mD·m, respectively, which suggests the "long fracture and high conductivity" fracturing strategy has been achieved.

Table 2—Fracture parameters evaluation for Stage 1-15

Stages No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	NOTE
Frac half length, m	108.6	108.5	108.7	108.5	108.3	116.8	116.8	116.6	116.6	116.7	123.5	123.5	123.6	123.4	123.4	Design
	106	94.8	95.4	94.4	95.8	102.1	100.6	101.9	103.7	107.2	111.9	109	90.1	105.2	104.7	Matching
Propped half Length, m	94.1	94.1	94.5	94	93.6	103.6	103.6	103.2	103.2	103.5	110.6	110.6	110.8	110.5	110.4	Design
	98.6	91.8	92.4	91.6	93	99.6	98	98.3	101.3	104.2	108	105.9	88.2	102.6	102.2	Matching
Avg. conc., kg/m ²	4.4	4.4	4.37	4.4	4.43	4.54	4.55	4.53	4.55	4.54	4.78	4.8	4.78	4.77	4.79	Design
	4.14	4.9	4.9	4.9	4.3	5.1	5.2	5.1	5.1	4.9	5.2	5.4	7	5.8	5.87	Matching
Avg. conductivity, mD·m	626.6	626	614.9	629.1	636.6	644.4	645.5	642.4	645.2	643.7	673.3	675.3	670.9	672	675.4	Design
	589.6	697.1	689.5	700.6	617.9	723.9	737.7	682.9	684.3	648.9	703	696	918	752	760	Matching
Max height, m	64.8	64.8	64.8	64.8	64.7	66.4	66.4	66.4	66.4	66.4	67.8	67.8	67.8	67.8	67.8	Design
	66.4	62.5	62	62.2	62.7	64	64.4	64.4	63.8	63.9	65.7	65.9	61	63.7	63.2	Matching
Fracture top height, m TVDSS	2752.7	2752.7	2752.7	2752.7	2752.8	2751.8	2751.8	2751.8	2751.8	2751.8	2750.9	2750.9	2750.9	2751.9	2750.9	Design
	2749.9	2752.5	2753.1	2752.8	2752.3	2751.4	2750.9	2750.9	2751.1	2750.8	2749.4	2750.1	2753.7	2750.9	2751.5	Matching
Fracture bottom height, m TVDSS	2817.5	2817.5	2817.5	2817.5	2817.5	2818.1	2818.2	2818.1	2818.1	2818.2	2818.7	2818.7	2818.7	2819.7	2818.7	Design
	2816.3	2815	2815.2	2815.1	2815	2815.4	2815.3	2815.3	2814.9	2814.7	2815	2815.9	2814.6	2814.7	2814.7	Matching

Table 3—Fracture parameters evaluation for Stage 16-29

Stages No.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	NOTE
Frac half length, m	123.5	123.5	123.3	123.4	123.6	123.4	123.4	123.9	124.1	123.2	129.4	129.4	129.4	129.5	Design
	102.4	112.3	108.2	108.7	109.3	106.8	108.7	109.2	112.9	108.8	113.4	113.1	113.2	114.1	Matching
Propped half Length, m	110.8	110.8	110.4	110.4	110.8	110.4	110.5	110.8	110.5	110.3	119.2	119.1	119.2	118.9	Design
	100.2	108.8	105.7	106.1	106.5	104.5	108.7	105.9	109.3	106.2	110.5	110.8	110.9	111.2	Matching
Avg. conc., kg/m ²	4.77	4.77	4.78	4.8	4.77	4.79	4.8	4.78	4.77	4.79	4.92	4.93	4.92	4.92	Design
	5.95	5.29	5.48	5.4	5.4	5.5	5.4	5.3	5.2	5.45	5.7	5.7	5.6	5.6	Matching
Avg. conductivity, mD·m	669.7	669.1	672.5	675.4	670.9	674.2	686.8	682.7	683.3	684.8	683.5	673.8	683.3	686.6	Design
	751.8	667.8	693.9	683.8	683.6	696.7	695.4	681.3	670.4	701.2	712.7	701.1	700.0	703.3	Matching
Max height, m	67.8	67.8	67.8	67.8	65.6	67.8	67.8	68	68	67.8	69	69	69	69	Design
	63.8	65.4	65.5	65.3	65.2	65.2	65.2	66	65.8	65.4	67	67.3	66.7	67.2	Matching
Fracture top height, m TVDSS	2750.9	2750.9	2750.9	2750.9	2750.4	2750.9	2750.9	2750.9	2750.8	2751	2750.1	2750.1	2750.1	2750.1	Design
	2751.4	2749.4	2750.3	2750.6	2750.5	2750.5	2749.6	2750	2749.2	2750.5	2749.1	2748.6	2749.4	2748.8	Matching
Fracture bottom height, m TVDSS	2818.7	2818.7	2818.7	2818.7	2816	2818.7	2818.7	2818.8	2818.9	2818.7	2819.1	2819.1	2819.1	2819.1	Design
	2815.2	2814.8	2815.9	2815.9	2815.7	2815.7	2814.8	2816	2815.1	2816	2816	2815.9	2816.1	2816.1	Matching

The 3D fracture profile model for 29 stages is simulated as shown in Fig. 10.

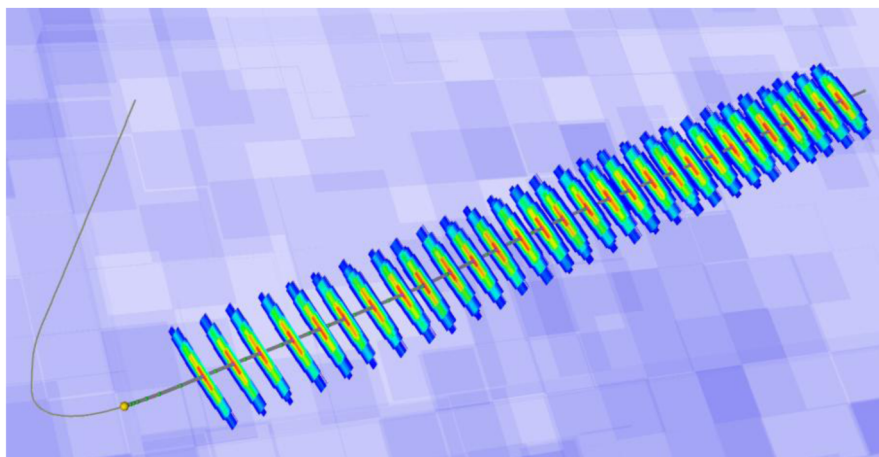


Figure 10—3D Fracture profile of well H1X (29 stages)

Post-frac production performance

After fracturing, H1X achieved a breakthrough production of 2,227 BOPD (28/64" choke). As of the latest data, the well produces at 1,850 BOPD with a 24/64" choke (Fig. 11). Its production index (~ 4.2 BOPD/psi) is 2–4 times higher than other open-hole fractured horizontal wells in the field, demonstrating the effectiveness of CT sand jetting combined with multi-stage proppant fracturing in unlocking ultra-low permeability carbonates.

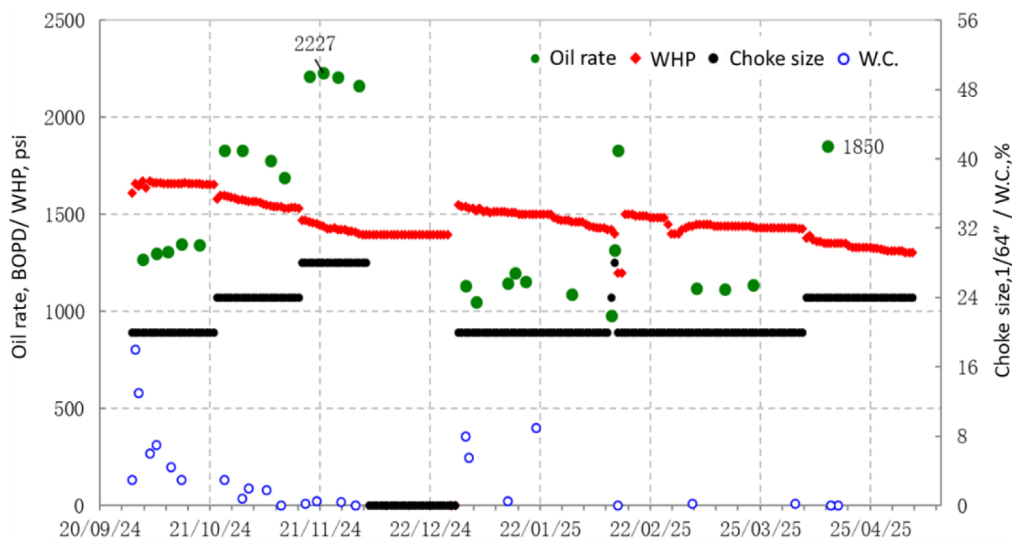


Figure 11—Production performance of well H1X

Conclusions

The successful application of the casing-cemented well integrating CT sand jetting with multi-stage proppant fracturing in the H1X pilot well demonstrates the technical viability of this approach for ultra-low permeability carbonate reservoirs. The holistic technical study based on "geology-engineering integration" ensures that each design step is optimized, leading to the excellent production performance of the well.

As the Middle East's first cased-hole multi-stage fracturing in ultra-low permeability carbonates, the H1X pilot well case study provides a replicable template for unlocking similar challenging reservoirs. This innovative approach combines CT sand jetting and multi-stage proppant fracturing, offering a new solution for the development of ultra-low permeability carbonate reservoirs in the region. The technical design, job

execution, and production performance data presented in this paper can serve as a valuable reference for future projects in similar geological settings.

Acknowledgments

The authors would like to express their gratitude to the shareholder Missan Oil Company(MOC, Iraq) and management of the Iraq's X Oilfield for their support and approval of this pilot project. Special thanks to the technical team involved in the geology evaluation, reservoir simulation, drilling, completion, and fracturing operations for their hard work and dedication. We also appreciate the assistance from the service companies in providing the necessary equipment and technical support.

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